

AJAY VENKATESH

Brooklyn, NY

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Education

New York University | Master of Science in Computer Engineering (GPA - 3.9/4)

Aug 2024 - Aug 2026

Experience

Amazon

May 2025 - Aug 2025

Software Development Engineer Intern

Seattle, US

- Designed a secure UI Console architecture using Midway SSO, OAuth, and HELIS to enforce least-privilege, role-based access; presented the solution to the team and received design approval.
- Built serverless infrastructure using AWS CDK to deploy API Gateway, Lambda (Java 21), DynamoDB, and S3 with cross-account access; implemented CI/CD pipeline for multi-stage deployments.
- Developed the frontend using React 18, integrating AWS UI components, Midway AuthN, and HELIS AuthZ with role-aware views and filtering; delivered a UI that was 40% faster and significantly more secure than older version.
- Implemented backend APIs using Java Coral, Smithy models, Dagger DI, and CBOR serialization; improved response time by 35% and ensured reliability through comprehensive unit and integration testing.

New York University

Aug 2024 - May 2025

Software Developer (On-Campus)

New York, US

- Contributed to an AI-driven Remote Patient Monitoring System, awarded \$100K by the National Science Foundation, USA, employing React Native, TypeScript, and Cognito authentication for real-time health monitoring.
- Authored parallelized ETL pipelines in Python to convert unstructured NoSQL data into structured SQL using GCP Cloud Functions. Leveraged concurrent processing of data chunks, reducing transformation time by 60%.
- Developed a React website to host QuickSight dashboard handling 50K+ records, utilized Cognito, Amplify, and Lambda functions to establish login, role-based access control, and Stripe integration for payment gateway.

PricewaterhouseCoopers Service Delivery Center (PwC SDC)

Aug 2021 - Aug 2024

Senior Software Engineer

- Built a distributed backend on Microsoft Azure for a health platform, integrating Microsoft Dynamics for secure data flow and using Azure services for scalability, fault tolerance, and event-driven processing.
- Led code reviews and mentored interns on design, TDD, and optimization, reducing bugs by 25% and accelerating CI/CD in an Agile setup.
- Developed a scalable B2B platform with modular services to replace legacy onboarding, integrating pipelines, email tracking, and BI reporting, improving automation by 70%.
- Deployed a cognitive neural model via REST APIs to extract structured data from physical cards using OCR and NLP, reducing manual effort by 75%.
- Implemented RBAC and session validation for 5+ roles, enforcing least-privilege access and preventing unauthorized actions.
- Optimized data pipeline by parallelizing Excel file imports (100K+ rows), cutting processing time from 5 hours to 1 and boosting efficiency by 80%.

Projects

Advanced Expense Tracker (Java Desktop Application) | Java, JavaFX, JDBC, SQLite, JFreeChart

- Architected a modular Java application supporting both individual and group expenses with secure login, leveraging OOP principles and design patterns (MVC and DAO) for separation of concerns.
- Implemented concurrent transaction handling and real-time cost splitting using multithreading and synchronized blocks to ensure thread-safe database updates in group expense scenarios.

Skills

Programming Languages: Python, Java, C++, JavaScript (ES6), SQL

Web Development: REST API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize

Database: RDBMS (MySQL, PostgreSQL), NoSQL (MongoDB, JSON), Dataverse

Cloud: AWS (AWS CDK, AWS SDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (AI services, Logic Apps, Security Data lakes, Function Apps, Storage blobs)

Tools: Git, Visual Studio, Jupyter, Postman, Docker, Power Automate, Power BI, QuickSight

Methodology: Agile, Waterfall, Scrum

AI/ML: Classification, Pattern Recognition, Data Extraction, MLOps, Neural Network, PCA, LLM

Certifications

AWS Cloud Practitioner

Microsoft Developer Associate

Power BI Data Analyst Associate

Microsoft Solution Architect Expert

Azure Data Scientist Associate

Azure Data Fundamentals

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (IEEE)

An Aquila-optimized SVM classifier for Diabetes prediction (IEEE)